

Spatial Memory & Local Web Search Canvas

"A browser extension that transforms tab management into an interactive 2D spatial canvas"

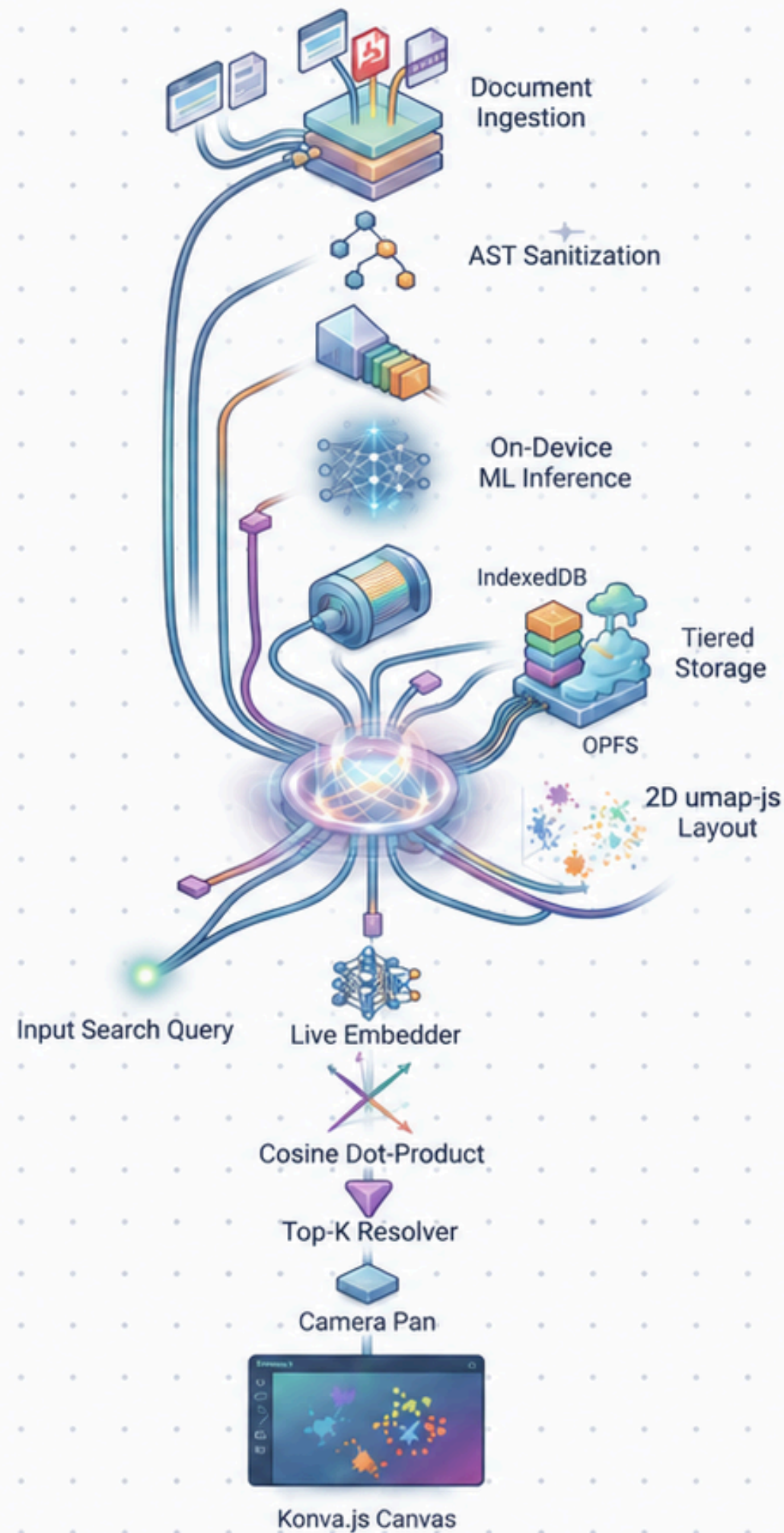
TEAM

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Submitted to: **Professor Jeelani Asif**



Introduction

Meridian: Latent Cartographer (our working name for the Spatial Canvas extension) turns your browser tabs into an interactive 2D map instead of a cramped strip of icons. It runs entirely on-device — a small WebAssembly AI model reads each page and quietly organises it for you, with nothing ever sent to a server.

- Automatically captures active tabs, PDFs and notes in the background
- Generates on-device semantic embeddings (Xenova/all-MiniLM-L6-v2)
- Groups related pages into visual clusters you can pan, zoom and explore
- 100% local-first — your browsing history never leaves your machine

384-d

on-device vector embeddings

<20 ms

search-by-intent latency

60 FPS

smooth, pannable spatial canvas

Motivation

Every developer and researcher knows this feeling: forty tabs open, three PDFs, half-finished searches — and no memory of where anything was. Regular tab bars and bookmarks were never built for this kind of research.

- Tab bars show only tiny, clipped titles — spatial memory (“it was on the left, near the blue diagram”) is completely lost
- Bookmark folders turn a living research trail into a flat, static list
- Cloud “smart search” tools send browsing history to a server, adding latency, cost and privacy risk
- Closing the laptop for the night shouldn’t mean losing a day of research context

“A zero-cost, local-first visual memory map lets you resume research with almost no friction — and total privacy.”

TODAY



Flat tabs — titles clipped, no memory of position

WITH SPATIAL CANVAS



Problem Statement

Four recurring pain points, straight from how developers and researchers actually work:

1 Fragmentation & Tab Overload

Information is scattered across flat, linear tab bars where titles and context get clipped.

2 Semantic Drift & Shallow Indexing

Bookmarks store static URLs; simple text search misses code snippets, tables and buried details.

3 Latency & Privacy Risk

Cloud-based “smart search” adds network lag, API cost, and exposes browsing history externally.

4 Memory Inefficiency

Unmanaged screenshots and raw page snapshots quietly exhaust browser memory and storage.

System Architecture

A decoupled, client-side Chromium extension (Manifest V3) — nothing ever leaves the device.

● Multi-source ingestion

Tabs, local PDFs (pdf.js), Markdown notes & code

● Local-first AI engine

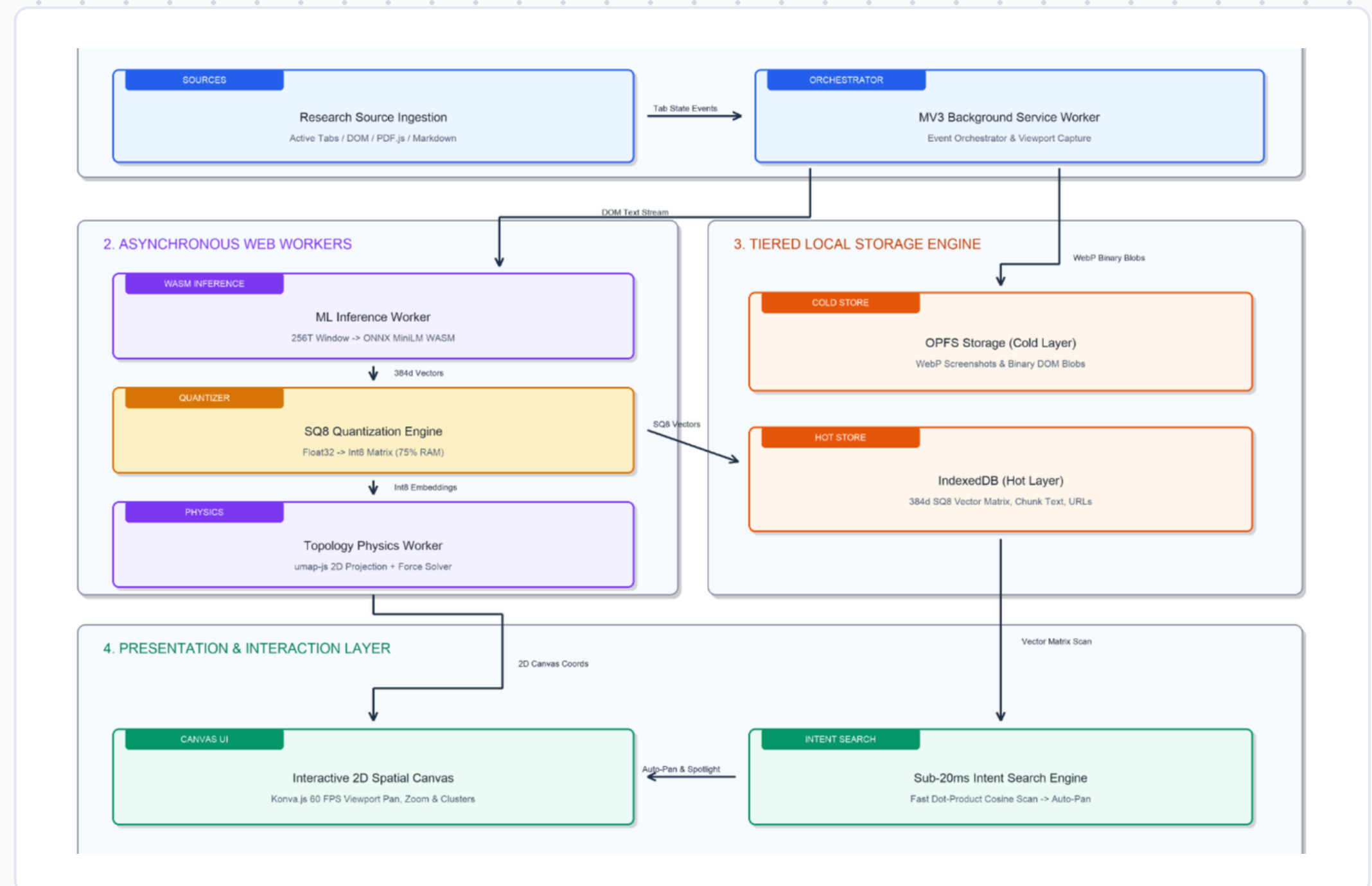
On-device MiniLM-L6-v2 via ONNX/WASM workers

● Interactive 2D canvas

Konva.js force-directed clusters at 60 FPS

● Tiered storage

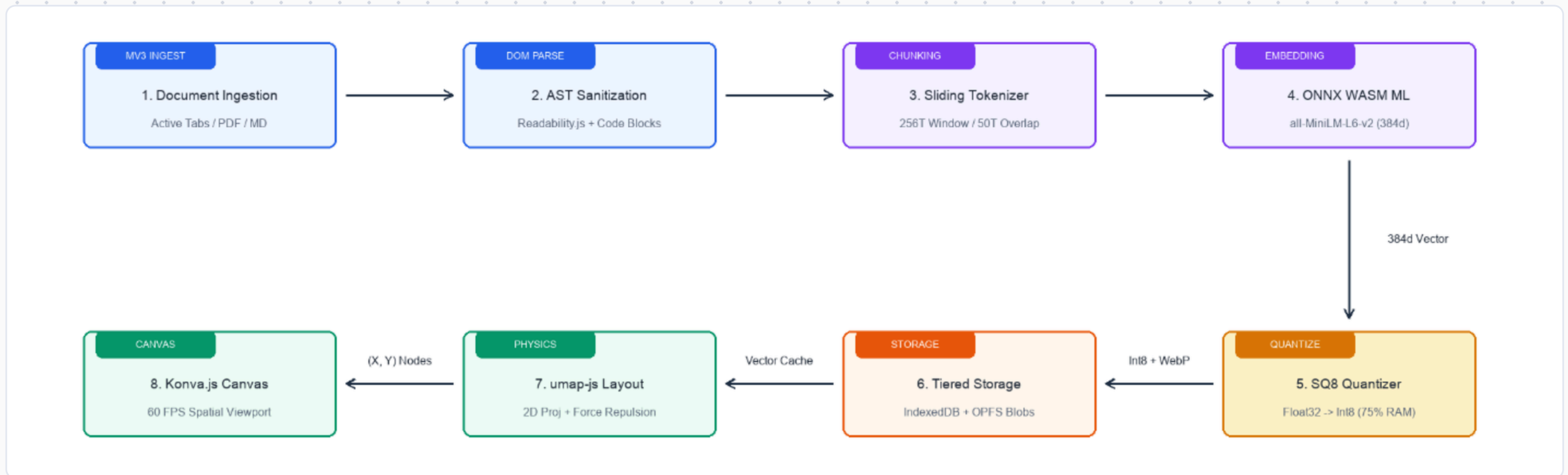
IndexedDB (hot) + OPFS (cold), SQ8-quantized



Meridian's end-to-end architecture — decoupled MV3 ingestion, async ML workers, tiered storage and the canvas UI.
(Source: team proposal, Fig. 3)

Capture & Embedding Pipeline

Every tab is pushed through an eight-stage pipeline — from raw DOM to a point on the canvas:



Squarish pipeline architecture — DOM sanitization, sliding-window chunking, on-device WASM embedding, SQ8 quantization and 60 FPS canvas projection. (Source: team proposal, Fig. 1)

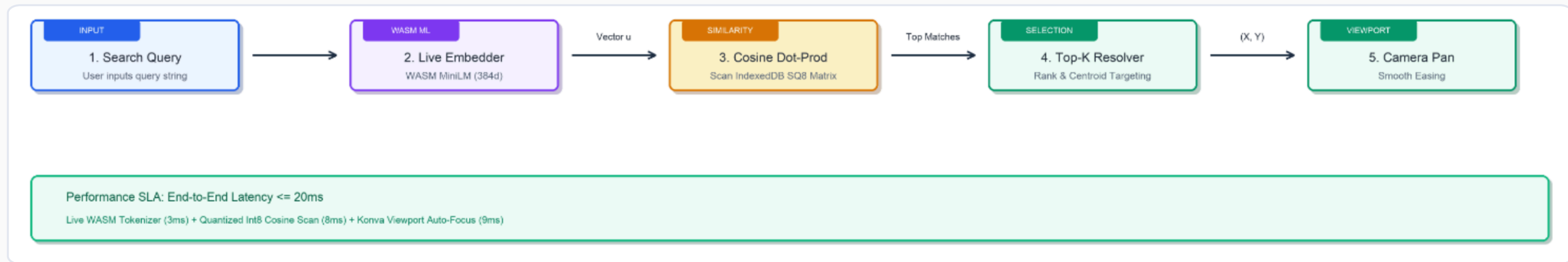
256-token window / 50-token overlap

384-d embeddings

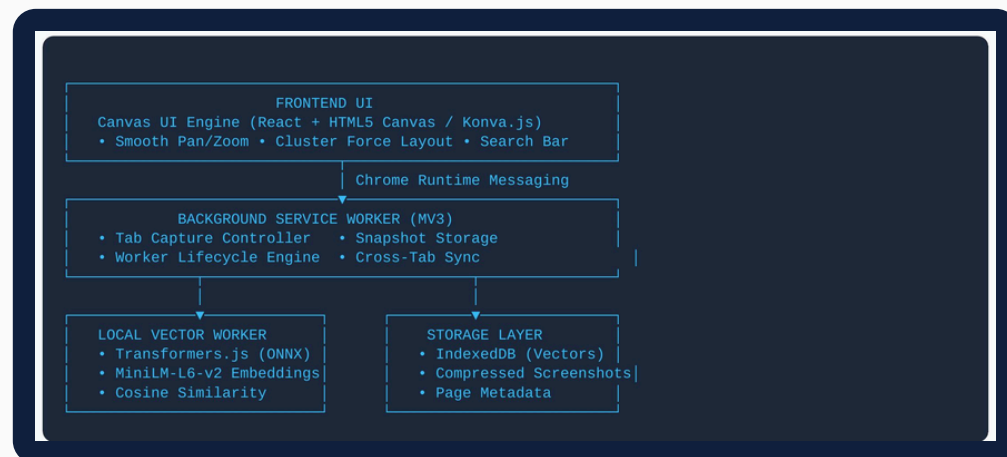
SQ8 → ~75% smaller

60 FPS force layout

Search, Storage & Delivery



Real-time search retrieval — live WASM embedding, SQ8 cosine scoring and smooth viewport easing, end to end in $\leq 20\text{ms}$. (Source: team proposal, Fig. 2)



System at a glance — Spatial Canvas PRD architecture diagram

BUILT WITH

React + Vite

Konva.js

Transformers.js (WASM)

IndexedDB + OPFS

Manifest V3

One click exports the whole canvas as a single, shareable .html file — vectors and thumbnails included, no backend required.